

Rahul Gupta

Senior Applied Science Manager | Responsible AI, Frontier Model Evaluation, ML Research Leadership

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EXECUTIVE PROFILE

- Research executive in Amazon's AGI organization leading Responsible AI for frontier models, including safety, red-teaming, evaluation science, fairness, privacy, and mitigation programs reported to VP- and SVP-level leaders.
- Built and led ML research organizations of 50+ people across Amazon Nova safety, LMArena performance, and the Nova Trusted AI Challenge; translate ambiguous frontier-model risks into measurable research agendas, product decisions, and external evaluation outcomes.
- Hands-on scientist with first-author work on [CBRN uplift evaluation for frontier language models](#) and [fairness in large-scale NLU systems](#); 25+ patented or patent-pending technologies and publications across NeurIPS, ICLR, ACL, EMNLP, FAccT, and ICASSP.

RESEARCH LEADERSHIP EXPERIENCE

Senior Applied Science Manager June 2026 – Present
Amazon / Frontier AI Assets, Responsible AI Cambridge, MA

- Overall head of Responsible AI in Amazon's newly architected AGI division, reporting directly to a Vice President.
- Set research and execution priorities for frontier model safety and evaluation programs spanning model capability assessment, policy alignment, red teaming, and launch readiness.

Senior Applied Science Manager October 2022 – May 2026
Amazon AGI / Nova Responsible AI Cambridge, MA

- Led Nova model safety work recognized by third-party evaluators [1], [2], [3].
- Led the Nova LMArena team, driving Nova models to top-15 LMArena performance by connecting evaluation diagnostics, post-training priorities, and model iteration.
- Founded and led the Nova Trusted AI Challenge, generating 100+ applications and selecting 10 participating university teams to advance external safety research and evaluation.

Applied Science Manager December 2018 – September 2022
Amazon Alexa / Responsible AI Cambridge, MA

- Founded and led a Responsible AI research team for Alexa, building foundational safety, fairness, and privacy capabilities into platform and product workflows.
- Drove research programs for private and unbiased customer experiences, with goals and outcomes reported to Amazon Senior Vice Presidents.

Applied Scientist August 2016 – November 2018
Amazon Alexa / Large-Scale NLU Cambridge, MA

- Served as technical lead on decoupling training and testing of large-scale ML systems.
- Delivered business impact that led to an organizational restructuring and influenced production NLU model development practices.

EDUCATION

PhD, Electrical Engineering, University of Southern California 2010 – 2016

Bachelor of Technology, Electrical Engineering, Indian Institute of Technology, Kharagpur 2006 – 2010

EXTERNAL RESEARCH IMPACT & SERVICE

- Organizer of the [TrustNLP](#) workshop series and workshops on trustworthy speech processing and LLM unlearning [1], [2]; reviewer for ACL, ICLR, ICASSP, and related top-tier venues.
- Point of contact for multiple [Amazon Academic Research Awards](#) collaborations, collectively representing over \$20M in funded research.
- Keynote speaker at venues including [IEEE SLT](#) and workshops on frontier AI advancement and social impact [1], [2].

SELECTED PUBLICATIONS

Full list available at my [Google Scholar page](#).

- [A Threshold Exceedance Framework for CBRN Uplift Evaluation in Frontier Language Models](#), arXiv preprint, 2026. First-author, hands-on experimentation.
- [Establishing Best Practices for Building Rigorous Agentic Benchmarks](#), NeurIPS 2025. Winner of the AgentX LLM Agents MOOC Competition, 2025.
- [FLIRT: Feedback Loop In-context Red Teaming](#), EMNLP 2024. Covered in the red teaming community [1], [2]; #3 paper of the day on Hugging Face.
- [“I’m fully who I am”: Towards centering transgender and non-binary voices to measure biases in open language generation](#), ACM FAccT 2023.
- [BOLD: Dataset and Metrics for Measuring Biases in Open-Ended Language Generation](#), ACM FAccT 2021. (700+ citations)

OTHER RECOGNITIONS

- Ming Hsieh Department of Electrical Engineering best PhD pitch award, 2015.
- Two-time winner of Interspeech computational paralinguistics challenge, 2013 and 2015.
- Phi Beta Kappa alumni in Southern California scholarship, 2015.
- Ming Hsieh Department of Electrical Engineering humanitarian award, 2014.
- Provost PhD fellowship, 2010–2014.